



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

10-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C	A,D	A,B,C	A,B,C	A,C	A,C,D	A,C	A,C,D	A,D	A,D
SECTION-III	Q.	1	2	3	4	5	6				
	A.	5	4	3	2	3	5				
SECTION-IV	Q.	1	2								
	A.	A->R,B->S,C->Q,D->P	A->R,B->Q,C->S,D->P								

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	B,C	A,B,D	A,B,C	B,C	A,B,C,D	A,B	A,B,C,D	B,C,D	A,C,D
SECTION-III	Q.	1	2	3	4	5	6				
	A.	9	9	5	3	3	1				
SECTION-IV	Q.	1	2								
	A.	A->PQR,B->S,C->PS,D->PT	A->RT,B->PQ,C->PQRS,D->QT								

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,D	B,D	B	A,B,C	B,C,D	A,B,C,D	A,B,C,D	A,C	B,C	B
SECTION-III	Q.	1	2	3	4	5	6				
	A.	1	3	1	2	5	5				
SECTION-IV	Q.	1	2								
	A.	A->Q,B->PQRS,C->T,D->Q	A->R,B->T,C->Q,D->Q								

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B,C)

$$H = 6 \left[KA \left(\frac{T_H - T_L}{x} \right) \right]$$

$$= 6 \times 10^{-2} \times 0.8 \times \left(\frac{30.0}{0.02} \right) = 72 \text{ J/s}$$

$$H = \left(\frac{dm}{dt} \right) L$$

$$\frac{dm}{dt} = \frac{H}{L} = \frac{72}{3.2 \times 10^5}$$

$$\text{Ice melted in 1 hour} = 3600 \frac{dm}{dt} = 810 \text{ gm}$$

2. Ans (A,D)

(A) We can easily see that charge q is placed symmetrically to surface ABCD, ABSR and ADQR. Charge q is also placed symmetrically to rest of the surfaces.

If the flux through the surface ABCD is x and through RSPQ is y then the total flux will be $3x + 3y$ Now by Gauss law

$$\Rightarrow 3x + 3y = \frac{q}{\epsilon_0}$$

$$\Rightarrow x + y = \frac{q}{3\epsilon_0}$$

(D) Flux through two surfaces are not same flux via ABCD is larger.

5. Ans (A,C)

$$-\frac{1}{F} = P = 2P_{I_1} + 2P_{I_2} + P_m \quad \dots(1)$$

$$P_{I_1} = \frac{1}{f_1} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$P_{I_1} = [(1.5 - 1)] \left[-\frac{1}{10} - \frac{1}{15} \right] = -\frac{1}{12} \quad \dots(2)$$

$$P_{I_2} = \frac{1}{f_2} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$P_{I_2} = \left(\frac{4}{3} - 1 \right) \left[\frac{2}{15} \right] = \frac{2}{45} \quad \dots(3)$$

$$P_m = -\frac{1}{f} = +\frac{2}{15} \quad \dots(4)$$

$$-\frac{1}{F} = P = 2 \left[-\frac{1}{12} + \frac{2}{45} \right] + \frac{2}{15} = -\frac{1}{6} + \frac{4}{45} + \frac{2}{15} = \frac{1}{8}$$

$F = -18\text{cm}$. Focus is negative means system will behave as concave mirror.

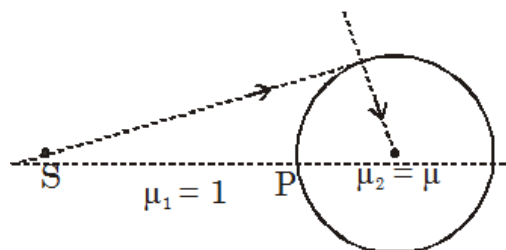
7. Ans (A,C)

$f = mg \sin \theta$ and $PE \sin \theta = mg \sin \theta$. R

$q2RE \sin \theta = mg \sin \theta R$

$$E = \frac{mg}{2q}$$

8. Ans (A,C,D)



$$(A) \frac{\mu}{+R} = \frac{\mu - 1}{+R} \text{ (not possible)}$$

The higher the refractive index of the material of sphere (μ), the closer one could come to fulfilling the objective.

$$(B) \frac{f_a}{f_w} = \left(\frac{\mu - \mu_w}{\mu - \mu_a} \right) \left(\frac{\mu_a}{\mu_w} \right)$$

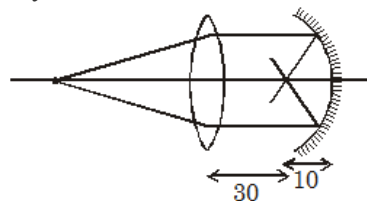
$$= \left(\frac{\frac{3}{2} - \frac{4}{3}}{\frac{3}{2} - 1} \right) \frac{1}{\left(\frac{4}{3} \right)} = \frac{1}{6} \times 2 \times \frac{3}{4} = \frac{1}{4}$$

(C) basic (NCERT point to ponder)

(D) basic (NCERT point to ponder)

9. Ans (A,D)

(A) Object for lens is between f & $2f$. Image is beyond $2f$.

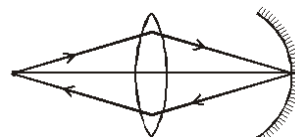


(B) For auto collimation obj for mirror should be at its C is at 20 cm from mirror.

In B case, image is virtual not possible

In C case, rays are parallel not possible

In D case, if object is at 40 cm image is formed at pole



10. Ans (A,D)

$$\frac{u_{\text{rel}}}{\sin 30^\circ} = \frac{u}{\sin \alpha}$$

$$u_{\text{rel}} = \frac{u \sin 30^\circ}{\sin \alpha}$$

$$u_{\text{rel}} = \text{min if } \alpha =$$

$$90^\circ$$

$$u_{\text{rel min}} = \frac{u}{2}$$

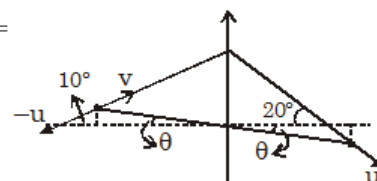


Figure (1)

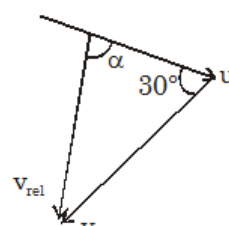


Figure (2)

$$v_{Ix} = +m^2 v_0 x$$

$$u \cos 20^\circ = m^2 \times v \cos 10^\circ$$

$$\text{also } v \cos 30^\circ = u$$

$$\frac{\cos 30 \cos 20}{\cos 10} = m^2$$

$$m^2 = \frac{[\cos 50 + \cos 10]}{2 \cos 10}$$

$$= \frac{\cos 50}{2 \cos 10} + 2$$

$$= \frac{\cos 60 \cos 10 + \sin 60 \sin 10}{2 \cos 10} + 2$$

$$= \frac{1}{4} + \frac{\sqrt{3}}{4} \times \frac{10\pi}{180} + 2$$

SECTION-III

1. Ans (5)

Let volume of glass vessel at 20°C is V_g and volume of mercury at 20°C is V_m

So volume of remaining space is $= V_g - V_m$

It is given constant so that

$$V_g - V_m = V'_g - V'_m$$

Where V'_0 and V'_m are final volumes.

$$V_g - V_m = V_g [1 + \gamma_g \Delta\theta] - V_m \{1 + \gamma_{Hg} \Delta\theta\}$$

$$\Rightarrow V_g \gamma_g = V_m \gamma_{Hg}$$

$$\Rightarrow V_m = \frac{100 \times 9 \times 10^{-6}}{1.8 \times 10^{-4}} \Rightarrow V_m = 50\text{cc.}$$

2. Ans (4)

$$\frac{\Delta Q}{\Delta t} = K_A A \frac{dT}{dx}$$

$$\frac{dT}{dx} \text{ in conductor A} = \text{slope of graph} = \sqrt{3}$$

Since both conductors are connected in series, same heat current will flow in A and B.

$$K_B = \frac{\Delta Q}{\Delta t} \times \frac{1}{A \frac{dT}{dn}} = \frac{\Delta\theta}{\Delta t} \times \frac{1}{A \tan 45^{\circ}}$$

$$= 200\sqrt{3}\text{J/m-sec }^{\circ}\text{C}$$

3. Ans (3)

$$\therefore \mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin\left(\frac{A+\frac{A}{2}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin\left(\frac{3A}{4}\right)}{\sin\left(\frac{A}{2}\right)}, \text{ Let } \frac{A}{4} = \theta$$

$$\Rightarrow \frac{2}{\sqrt{3}} = \frac{\sin 3\theta}{\sin 2\theta} = \frac{3 \sin \theta - 4 \sin^3 \theta}{2 \sin \theta \cdot \cos \theta} = \frac{3 - 4 \sin^2 \theta}{2 \cos \theta}$$

$$\Rightarrow 4\sqrt{3}\cos^2 \theta - 4 \cos \theta - \sqrt{3} = 0$$

$$\Rightarrow \cos \theta = \frac{1 \pm 2}{2\sqrt{3}} = \frac{\sqrt{3}}{2}, \left(-\frac{1}{2\sqrt{3}}\right)$$

$$\Rightarrow \theta = 30^{\circ} \Rightarrow \frac{A}{4} = 30^{\circ} = \frac{\pi}{6}$$

$$\Rightarrow A = \frac{2\pi}{3}, \therefore C = 3$$

4. Ans (2)

The minima will be heard at P when a creast from S_1 and a trough from S_2 reach there at the same time. This will happen if $L_1 - L_2$ is $\frac{\lambda}{2}$

or $\lambda + \left(\frac{\lambda}{2}\right)$ or $2\lambda + \left(\frac{\lambda}{2}\right)$ and so on. Hence,

the increase in L_1 between consecutive minima is λ_1 and from the data we see that $\lambda = 0.40\text{ m}$.

$$\text{Then } \lambda = \frac{v}{f} \Rightarrow f = \frac{340}{0.40} = 850\text{Hz.}$$

5. Ans (3)



$$P = 2\sigma AT_1^4 - \sigma AT_2^4 \text{ (steady-state condition of sheet I)(i)}$$

$$\text{and } 2\sigma AT_2^4 = \sigma AT_1^4 \text{ (steady state condition of sheet II)(ii)}$$

$$\Rightarrow P = 3\sigma AT_2^4 = 3\text{kW}$$

6. Ans (5)

$$2(1.5 - 1) 2 \left(\frac{1}{-R} - \frac{1}{R} \right) = P$$

$$-\frac{2}{2R} = P$$

$$\frac{-2}{R} + (\mu - 1) \left(\frac{1}{R} + \frac{1}{R} \right) = \frac{P}{3}$$

$$\frac{2}{R} (\mu - 2) = \frac{P}{3} = \frac{-2}{3R}$$

$$\mu - 2 = \frac{-1}{3}$$

$$\mu = 2 - \frac{1}{3} = \frac{5}{3}$$

PART-2 : CHEMISTRY

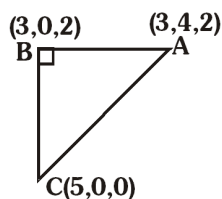
SECTION-I

3. Ans (A,B,D)

$$W = \frac{M/n_f}{F} \times \eta \times i \times t$$

4. Ans (A,B,C)

$AB \perp BC$



$\Rightarrow \triangle ABC$ right angle at B

$\therefore B$ is orthocentre

(A) $OB = \sqrt{13}$

(B) Circumcentre is mid-point of AC i.e. (4,2,1)

(C) If mid-point of BD & AC is same.

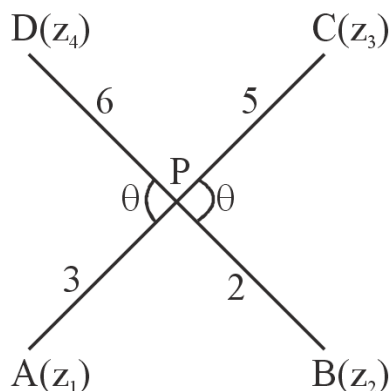
5. Ans (B,C,D)

Let A, B, C, D denote the complex numbers z_1, z_2, z_3 & z_4 respectively in the argand plane.

$$\text{Now } \frac{5z_1 + 3z_3}{8} = \frac{6z_2 + 2z_4}{8}$$

P divides AC in the ratio 3 : 5

& P divides BD in the ratio 1 : 3.



Let $AP = 3\lambda$, $CP = 5\lambda$

$BP = k$, $DP = 3k$

then in $\triangle APD$ $AD^2 = AP^2 + DP^2 - 2(AP)(DP)\cos\theta$

$$\Rightarrow |z_1 - z_4|^2 = 9\lambda^2 + 9k^2 - 2(3\lambda)(3k)\cos\theta$$

in $\triangle PBC \Rightarrow |z_2 - z_3|^2 = 25\lambda^2 + k^2 - 2(5\lambda)(k)\cos\theta$

$$\text{Given } 5|z_1 - z_4|^2 = 9|z_2 - z_3|^2$$

$$\Rightarrow 5 \times 9(\lambda^2 + k^2) = 9(25\lambda^2 + k^2) \Rightarrow k^2 = 5\lambda^2$$

$\therefore AP \times PC = BP \times PD \Rightarrow A, B, C, D$

are concyclic.

6. Ans (A,B,C,D)

$$k = |a + b\omega + c\omega^2| + |a + b\omega^2 + c\omega|$$

$$k = 2|a + b\omega + c\omega^2|$$

$$k^2 = 4(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$$

$$k^2 = 4(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= 2((a-b)^2 + (b-c)^2 + (c-a)^2)$$

clearly k is minimum when a, b and c are consecutive integers

$$\therefore k_{\min} = 2\sqrt{3}$$

7. Ans (A,B,C,D)

$$P(E_1) = \frac{1}{15} \times \frac{10}{14} \times \frac{9}{13} \times 3 = \frac{9}{91}$$

$$P(E_2) = 3 \times \frac{1}{15} \times \frac{9}{14} \times \frac{8}{13} = \frac{36}{455}$$

$$P(E_1 \cap E_2) = \frac{1}{15} \times \frac{1}{14} \times \frac{4}{13} \times 3 = \frac{4}{455}$$

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\frac{4}{455}}{\frac{36}{455}} = \frac{1}{9}$$

8. Ans (A,C)

$$P = \frac{{}^nC_0 \cdot {}^nC_0}{2^n \cdot 2^n} + \frac{{}^nC_1 \cdot {}^nC_1}{2^n \cdot 2^n} + \frac{{}^nC_2 \cdot {}^nC_2}{2^n \cdot 2^n} + \dots + \frac{{}^nC_n \cdot {}^nC_n}{2^n \cdot 2^n}$$

$$P = \frac{C_0^2 + C_1^2 + \dots + C_n^2}{2^{2n}} = \frac{{}^{2n}C_n}{2^{2n}}$$

$$P = \frac{|2n|}{2^{2n} |n| |n|} = \frac{1.2.3.4.5 \dots (2n-1)(2n)}{2^n (|n|)^2}$$

$$= \frac{(2.4.6 \dots 2n)(1.3.5 \dots (2n-1))}{2^{2n} (|n|)^2}$$

$$= \frac{(1.3.5 \dots (2n-1))}{2^n |n|}$$

9. Ans (B,C)

$$A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

$$A^3 = \begin{bmatrix} \cos 3\alpha & \sin 3\alpha \\ -\sin 3\alpha & \cos 3\alpha \end{bmatrix},$$

$$A^4 = \begin{bmatrix} \cos 4\alpha & \sin 4\alpha \\ -\sin 4\alpha & \cos 4\alpha \end{bmatrix}$$

$$B = A + A^2 + A^3 + A^4$$

$$B = \begin{bmatrix} \cos \alpha + \cos 2\alpha + \cos 3\alpha + \cos 4\alpha & \sin \alpha + \sin 2\alpha + \sin 3\alpha + \sin 4\alpha \\ [0.9em] -(\sin \alpha + \dots + \sin 4\alpha) & \cos \alpha + \dots + \cos 4\alpha \end{bmatrix}$$

$$\therefore \cos \frac{\pi}{5} + \cos \frac{2\pi}{5} + \cos \frac{3\pi}{5} + \cos \frac{4\pi}{5}$$

$$= \frac{\sin(4 \cdot \frac{\pi}{10})}{\sin \frac{\pi}{10}} \cdot \cos\left(\frac{\pi}{2}\right) = 0$$

$$\sin \frac{\pi}{5} + \sin \frac{2\pi}{5} + \sin \frac{3\pi}{5} + \sin \frac{4\pi}{5} = \frac{\sin(\frac{2\pi}{5})}{\sin \frac{\pi}{10}} \cdot 1$$

$$|B| = \frac{\sin^2 \frac{2\pi}{5}}{\sin^2 \frac{\pi}{10}} = \frac{(\sin 72^\circ)^2}{(\sin 18^\circ)^2} = (\cot 18^\circ)^2$$

10. Ans (B)

The coin can turn up heads 0,2,4,...,50 times to satisfy the condition.

Hence probability is :

$$P = {}^{50}C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^{50} + {}^{50}C_2 \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^{48} + \dots + {}^{50}C_{50} \left(\frac{2}{3}\right)^{50}$$

$$= \frac{\left(\frac{1}{3} + \frac{2}{3}\right)^{50} + \left(\frac{1}{3} - \frac{2}{3}\right)^{50}}{2} = \frac{3^{50} + 1}{2 \cdot 3^{50}}$$

SECTION-III

1. Ans (1)

$$\text{Let } C = 1 + \frac{\cos \theta}{5} + \frac{\cos 2\theta}{5^2} + \frac{\cos 3\theta}{5^3} + \dots$$

$$\& S = \frac{\sin \theta}{5} + \frac{\sin 2\theta}{5^2} + \frac{\sin 3\theta}{5^3} + \dots$$

$$\therefore C + iS = 1 + \frac{e^{i\theta}}{5} + \frac{e^{2i\theta}}{5^2} + \frac{e^{3i\theta}}{5^3} + \dots$$

$$= \frac{1}{1 - \frac{e^{i\theta}}{5}} = \frac{5}{5 - \cos \theta - i \sin \theta}$$

comparing real parts

$$\therefore C = \frac{5(5 - \cos \theta)}{(5 - \cos \theta)^2 + \sin^2 \theta}$$

$$= \frac{5(5 - \frac{1}{3})}{(5 - \frac{1}{3})^2 + (\frac{2\sqrt{2}}{3})^2} = \frac{105}{102}$$

$$\Rightarrow \left[\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{\cos k\theta}{5^k} \right] = \left[\frac{105}{102} \right] = 1$$

2. Ans (3)

$$\hat{p} \times 2\hat{q} - \vec{p} \times 2\hat{r} = 2\hat{q} - 2\hat{r}$$

$$\Rightarrow \vec{p} \times (2\vec{q} - 2\vec{r}) = (2\vec{q} - 2\vec{r})$$

which is possible only if $2\vec{q} - 2\vec{r} = \vec{0}$ i.e.

$$\vec{q} = \vec{r}$$

$$\therefore |3\hat{p} + 2\hat{q} - 2\hat{r}| = |3\hat{p}| = 3$$

3. Ans (1)

$$\text{Volume} = \left| \begin{array}{ccc} (4-1) & (6-3) & (12+1) \\ 2 & 2 & 1 \\ 2 & -1 & 1 \\ 3 & 4 & 12 \end{array} \right| = \frac{13}{7}$$

4. Ans (2)

$$\begin{aligned}\text{Clearly } F(F(z)) &= \frac{\frac{z+i}{z-i} + i}{\frac{z+i}{z-i} - i} \\ &= \frac{z + iz + i + 1}{z + i - iz - 1} \\ &= \frac{(z+1)(1+i)}{(z-1)(1-i)} \\ &= i \left(\frac{z+1}{z-1} \right)\end{aligned}$$

$$\text{Also } f(f(f(z))) = i \left(\frac{\frac{z+i}{z-i} + 1}{\frac{z+i}{z-i} - 1} \right) = i \frac{2z}{2i} = z$$

$$\therefore z_{2020} = z_1$$

$$\text{Now } z_1 = \frac{2+2i}{2} = 1+i$$

5. Ans (5)

$$\vec{a} \times \vec{u} = \vec{a} \times \vec{b}$$

$$\vec{a} \times (\vec{a} \times \vec{u}) = \vec{a} \times (\vec{a} \times \vec{b})$$

$$(\vec{a} \cdot \vec{u}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{u} = (\vec{a} \cdot \vec{b}) \vec{a} - (\vec{a} \cdot \vec{a}) \vec{b}$$

$$-14\vec{u} = 7\vec{a} - 14\vec{b}$$

$$-2\vec{u} = (\hat{i} + 2\hat{j} - 3\hat{k}) - 2(2\hat{i} + \hat{j} - \hat{k})$$

$$= -3\hat{i} - \hat{k}$$

$$\vec{u} = \frac{3}{2}\hat{i} + \frac{\hat{k}}{2}$$

$$|\vec{u}| = \sqrt{\frac{9}{4} + \frac{1}{4}} = \sqrt{\frac{5}{2}}$$

$$\frac{|\vec{u}|^2}{2} = \frac{5}{4} = 1.25$$

6. Ans (5)

No. of 'f' function

$$= 3^4 - [{}^3C_1 2^4 - {}^3C_2 (1)^4 + {}^3C_3 (0)^4]$$

$$= 81 - [48 - 3]$$

$$= 36$$

Now $f(a_1) = x$ and $f(a_2) = y$

$f(a_3) = z$ and $f(a_4) = x$ or $y = 2$

or

$f(a_3) = z$ and $f(a_4) = z = 1$

No. of 'f' functions = 5

$$\text{Probability} = \frac{5}{36} = .138 \text{ or } .139$$

SECTION-IV

1. Ans (A→Q, B→PQRS, C→T, D→Q)

$$|2z + 3i| = |z|^2$$

$$\therefore |2z + 3i| \leq 2|z| + 3$$

$$\Rightarrow |z|^2 \leq 2|z| + 3$$

$$\Rightarrow |z|^2 - 2|z| - 3 \leq 0$$

$$\Rightarrow 0 \leq |z| \leq 3$$

$$\text{again } |2z + 3i| \geq |2z| - 3.$$

$$\Rightarrow |z|^2 \geq |2z| - 3$$

Solving which, we get $|z| \geq 1$

$$\therefore 1 \leq |z| \leq 3$$

$$(A) |z|_{\max} = 3$$

$$(B) |z|_{\min} = 1$$

$$(C) \therefore |z_1 + z_2| = |z_1| + |z_2|$$

$\Rightarrow O, z_1, z_2$ are collinear with z_1, z_2 on same side of O.

$\therefore |2z + 3i| = 2|z| + |3i|$ at O, $2z, 3i$ collinear

$$\Rightarrow \text{for } |z|_{\max} : z = 3i$$

$$\therefore \alpha = 0, \beta = 3 \Rightarrow \alpha^3 + \beta^3 = 27$$

$$(D) \therefore |z_1 + z_2| = ||z_1| - |z_2||$$

$$\Rightarrow z_1, z_2, O \text{ collinear}$$

with z_1, z_2 on opposite side of O.

$$\therefore |2z + 3i| = ||2z| - |3i||$$

$$\text{for } |z|_{\min} : z = -i$$

$$\therefore x = 0, y = -1 \Rightarrow x^2 + 2y^2 = 2$$

2. Ans (A→R, B→T, C→Q, D→Q)

$$(A) \quad n(S) = \underline{5}, \quad n(E) = 1 + {}^5C_2 + \frac{{}^5C_2 {}^3C_2}{\underline{2}} = 26$$

$$\text{Required probability} = \frac{26}{120} = \frac{13}{60}$$

(B)

$$\sum_{k=0}^{50} \omega^{kl} = \omega^{0l} + \omega^{1l} + \omega^{2l} + \underbrace{\omega^{3l} + \omega^{4l} + \dots + \omega^{50l}}_{48 \text{ terms all } 1}$$

$$= \omega^1 + \omega^1 + \omega^2 + 48 \times 1$$

$$= \omega - 1 + 48 = \omega + 47$$

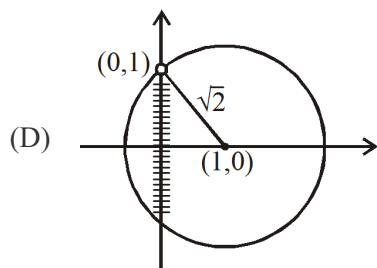
$$\lambda = 47$$

(C) $a = b = c \geq 1$

$$\text{Also, } 3a < 12$$

$$\Rightarrow 1 \leq a = b = c \leq 3$$

$$\text{Required probability} = \frac{3}{6} = \frac{1}{2}$$



$$\arg\left(\frac{z-i}{i}\right) = \pi \Rightarrow \arg(z-i) - \arg(i) = \pi$$

$$\Rightarrow \arg(z-i) = \frac{3\pi}{2} \text{ it is a ray emanating}$$

from (0, 1) along negative imaginary axes.

Now, $z\bar{z} - z - \bar{z} = 1$ it is circle with centre at (0,0) and radius $\sqrt{2}$ units.

$$\Rightarrow \text{required length} = 2.$$



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JEE (Advanced)

REVIEW TEST

10-10-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	A,C,D	A,B,C,D	A,B,C,D	A,B,C	B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A,C	A,B,C	A,C,D	A,C,D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	0.30	4.00	8.00	10.00	1.00	5.00	0.41 to 0.42

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C,D	A,B,C	B,C,D	A,B,D	B,C,D	D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	B	D	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	4.00	5.00	6.00	4.00	5.00	7.00	9.87

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B	A,C,D	A,B,D	B,D	B,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	C	A,B,D	A,C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.66 or 0.67	31.00	64.00	99.00	5.83	162.00	3.00	0.37 or 0.38

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,B,D)

By first law of thermodynamics

$$Q = W + \Delta U$$

$$\Rightarrow 2Q = \Delta U \dots (i)$$

$$\Rightarrow 2nC (T_B - T_A) = n \frac{5R}{2} (T_B - T_A)$$

$$\therefore C = \frac{5R}{4}$$

By (i)

$$Q = \frac{1}{2} \Delta U = \frac{1}{2} \left[\frac{5}{2} (6P_0 V_0 - 4P_0 V_0) \right] = \frac{5}{2} P_0 V_0$$

since $\Delta U = -2W$, therefore temperature goes on increasing from A to B

2. Ans (A,C,D)

Apply Gauss law and property of conductor.

3. Ans (A,B,C,D)

$y = 2a \cos (\omega t + \phi) \cos (Kx)$ comparison with given equation gives

$$K = \frac{2\pi}{\lambda} = 10\pi \Rightarrow \lambda = \frac{1}{5} \text{ m}$$

$$\omega = 2\pi f = 50\pi \Rightarrow f = 25 \text{ Hz}$$

$$\therefore v = 5 \text{ m/s}^{-1}$$

$$\text{at } x = 0.15 \text{ m}$$

$$\cos (10 \pi \times 0.15) = \cos (1.5 \pi) = 0 \text{ for all } t$$

$$\text{at } x = 0.3$$

$$\cos (10 \pi \times 0.3) = \cos 3\pi = -1 \text{ for all } t$$

4. Ans (A,B,C,D)

Since both the gases are contained in the same vessel, temperature of both the gases is same. Average KE per molecule of a diatomic gas is $\frac{5}{2} KT$. Hence, average KE per molecule of both the gases is same. Therefore, option (a) is correct.

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$\text{Hence, } \frac{(v_{rms})_2}{(v_{rms})_1} = \sqrt{\frac{M_1}{M_2}} = \sqrt{16} = 4$$

Hence, option(b) is correct.

Let molar mass of B be M , then that of A will be equal to $16M$.

Let mass of gas B in the vessel be m , then that of A will be $2m$. The number of moles of a gas, in the vessel will be $n = m/M$. hence, number of moles of gases A and B will be

$$n_1 = \frac{2m}{16M} \text{ and } n_2 = \frac{m}{M}$$

$$\text{Hence, } \frac{n_1}{n_2} = \frac{1}{8}$$

Hence, option (d) is correct

Partial pressure exerted by a gas is

$$P = \frac{nRT}{V}$$

$$\text{Hence, } \frac{P_2}{P_1} = \frac{n_2}{n_1} = 8$$

5. Ans (A,B,C)

$$e = a$$

$$\text{And } \rho = \frac{m}{v}$$

SECTION-I (ii)

8. Ans (A,B,C)

$$\frac{1}{2}mv^2 - \frac{Ze^2}{4\pi\epsilon_0 r_0} = 0$$

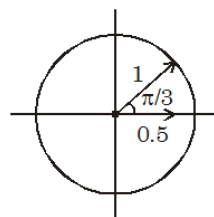
$$v = \sqrt{\frac{2Ze^2}{4\pi m\epsilon_0 r_0}}$$

$$3 \times 10^8 = \sqrt{\frac{2 \times Z \times 2.56 \times 10^{-38} \times 9 \times 10^9}{r_0 \times 9 \times 10^{-31}}}$$

$$9 \times 10^{16} < 200 \times 2.56 \frac{Z}{r_0}$$

$$\frac{Z}{r_0} < \frac{9 \times 10^{16}}{512}$$

9. Ans (A,C,D)



$$\frac{\pi}{3} = \omega t = 2\pi \times 400t$$

$$t = \frac{1}{2400} \text{ sec}$$

$$P = +ve \Rightarrow \vec{F} \cdot \vec{v} > 0$$

$$F_v \cos 0^\circ > 0$$

$$\text{Since } v \text{ is } +x \text{ at } t = 0$$

$$v = \sqrt{\frac{N}{P}} = f\lambda$$

$$\lambda = \frac{1}{400} \sqrt{\frac{10^{11}}{7840}} = \frac{10^5}{400 \times 28} = \frac{125}{14}$$

10. Ans (A,C,D)

$$(A) p = B \frac{\partial s}{\partial x}$$

$$v_p = -v \frac{\partial s}{\partial x}$$

$$\frac{v_p}{p} = \frac{v}{B}$$

$$\text{if } v_p > 0 \Rightarrow p > 0$$

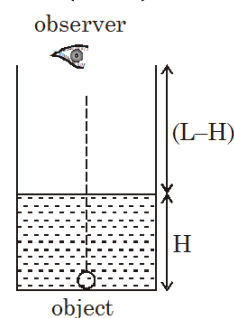
(B) & (C) at mean PE is maximum

$$v_{max} = A\omega$$

$$= 10^{-3} \times 2\pi \times 400 = \frac{4\pi}{5} \text{ m/s}$$

SECTION-II

1. Ans (5.00)



$$X_{app} = L - H + \frac{H}{\mu}$$

$$\frac{dX_{app}}{dt} = \frac{dH}{dt} \left(\frac{1-\mu}{\mu} \right)$$

$$\pi r^2 H = V$$

$$\therefore \frac{dH}{dt} = -\frac{2H}{r} \frac{dr}{dt} = \frac{2KH}{r}$$

$$\therefore \frac{dX_{app}}{dt} = \frac{2KV}{\pi r^3} \left(\frac{1-\mu}{\mu} \right)$$

2. **Ans (0.30)**

$$V_A = V_C$$

$$c = a + b$$

3. **Ans (4.00)**

Consider three more identical surfaces, such that a square shaped cylinder is formed with point charge at the centre. Flux will be equally divided between these surfaces.

4. **Ans (8.00)**

$$\frac{4\ell_1}{3} = \frac{2\ell_2}{4}$$

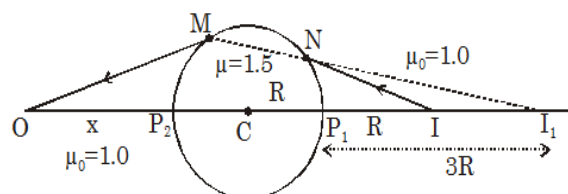
7. **Ans (5.00)**

We can reverse the question. Let I behaves as object and O behaves as image, because retracing the path will not alter the distances.

Consider first refraction at point N

$$\frac{1.5}{v} - \frac{1.0}{-R} = \frac{1.5 - 1.0}{+R}$$

$$\Rightarrow \frac{1.5}{v} - \frac{1}{2R} - \frac{1}{R} \Rightarrow v = -3R$$



Consider second refraction at point M, for this

I, will behaves as normal object so

$$\frac{1.0}{v} - \frac{1.5}{-5R} = \frac{1.0 - 1.5}{-R}$$

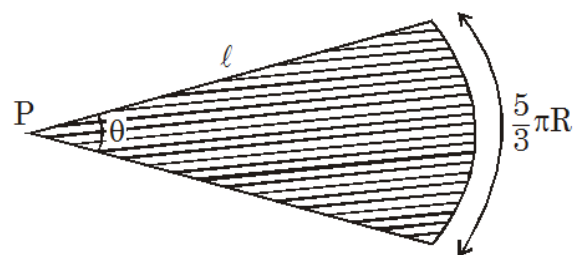
$$\Rightarrow \frac{1}{v} + \frac{3}{10R} = \frac{1}{2R} = \frac{5}{10R}$$

$$\Rightarrow \frac{1}{v} = \frac{5}{10R} - \frac{3}{10R} = \frac{1}{5R}$$

$$\Rightarrow v = 5R$$

$$\Rightarrow x = 5R \Rightarrow k = 5$$

8. **Ans (0.41 to 0.42)**



Potential due to circular non conducting disc,

$$V_C = \frac{\sigma r}{2\epsilon_0} \text{ (where r is radius of disc)}$$

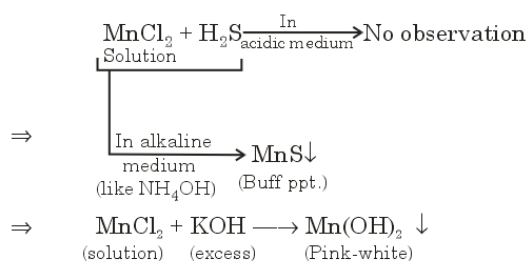
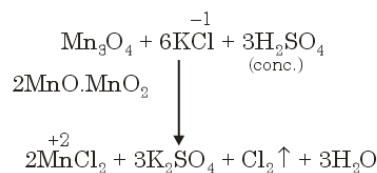
$$\text{So, } v_P = \left(\frac{\theta}{2\pi} \right) \left(\frac{\sigma \cdot \ell}{2\epsilon_0} \right)$$

$$\Rightarrow \left(\frac{\sigma \ell}{2\epsilon_0} \right) \left(\frac{5\pi R}{3(\ell)} \right) \left(\frac{1}{2\pi} \right) = \frac{5\sigma R}{12\epsilon_0}$$

PART-2 : CHEMISTRY

SECTION-I (i)

4. **Ans (A,B,D)**



SECTION-II

1. **Ans (5.00)**

$$\rho = \frac{2 \times 96 / N_A}{(4 \times 10^{-8} \text{ cm})^3} = 5 \text{ gm/cc}$$

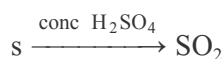
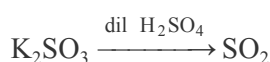
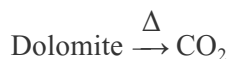
2. **Ans (4.00)**

$$E = E^\circ - 0.06 \text{ pH}$$

$$0.22 = 0.46 - 0.06 \text{ pH}$$

$$0.06 \text{ pH} = 0.46 - 0.22$$

6. **Ans (5.00)**



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,B)**

$$\text{It is possible } |\vec{a} \times \vec{b}| = 1 = |\vec{c} \times \vec{d}| \quad \&$$

$$\vec{a} \times \vec{b} \parallel \vec{c} \times \vec{d}$$

$$\therefore \vec{a} \perp \vec{b}, \vec{c} \perp \vec{d} \text{ \& } (\vec{a} \times \vec{b}) \parallel (\vec{c} \times \vec{d})$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$= k(\vec{c} \times \vec{d}) \cdot \vec{c} = 0$$

$$\Rightarrow [\vec{a} \ \vec{b} \ \vec{c}] = 0$$

2. **Ans (A,C,D)**

$$(A) \text{ Number of location} = {}^5C_3$$

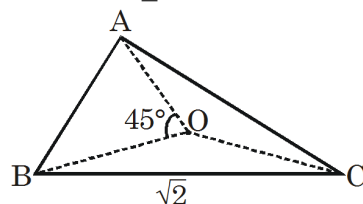
$$(C) \text{ Number of location} = {}^{10}C_3$$

$$(D) \text{ Number of location} = {}^{10}C_3 - {}^6C_3$$

3. **Ans (A,B,D)**

$$\therefore OA + OB = 4$$

$$\Rightarrow OA \cdot OB \leq 4$$



$$\therefore \text{ar}(\Delta AOB) = \frac{1}{2}(OA)(OB) \times \frac{1}{\sqrt{2}} \leq \sqrt{2}$$

Let h = altitude drawn from the vertex C to the base AOB.

$$\text{So, } h \leq \sqrt{2}$$

$$\therefore V = \frac{1}{3} \times \text{ar}(\Delta AOB) \times h \leq \frac{1}{3} \times \sqrt{2} \times \sqrt{2} = \frac{2}{3}$$

$$\text{But } V = \frac{2}{3} \Rightarrow \text{equality holds everywhere}$$

$$\text{So, } OA = OB = 2 \text{ \& } h = \sqrt{2}$$

$$\text{Also, } BC \perp \text{base AOB} \Rightarrow \angle OBC = 90^\circ$$

$$\text{So, } OC = \sqrt{6}$$

$$\text{And, } AB = \sqrt{4 + 4 - 8 \times \frac{1}{\sqrt{2}}} = \sqrt{8 - 4\sqrt{2}}$$

$$\therefore AC = \sqrt{8 - 4\sqrt{2} + 2} = \sqrt{10 - 4\sqrt{2}}$$

$$> \sqrt{9 - 4\sqrt{2}} = (2\sqrt{2} - 1)$$

4. **Ans (B,D)**

$$\text{tr}((A - AB)(A^T - B^T A^T)) = 0$$

$$\Rightarrow A - AB = O_n$$

$$\text{Since, } \text{tr}(CC^T) = 0$$

$$\Rightarrow C = O_n$$

5. **Ans (B,D)**

$$\vec{n}_1 = 2\hat{i} - 2\hat{j} - \hat{k} \Rightarrow \hat{n}_1 = \frac{2\hat{i} - 2\hat{j} - \hat{k}}{3}$$

$$\vec{n}_2 = 8\hat{i} + \hat{j} - 4\hat{k} \Rightarrow \hat{n}_2 = \frac{8\hat{i} + \hat{j} - 4\hat{k}}{9}$$

$$\hat{n}_1 + \hat{n}_2 = \frac{14\hat{i} - 5\hat{j} - 7\hat{k}}{9}$$

$$\hat{n}_1 - \hat{n}_2 = \frac{-2\hat{i} - 7\hat{j} + \hat{k}}{9}$$

$\therefore$ Angle bisector of lines are parallel to $(\hat{n}_1 + \hat{n}_2)$ and $(\hat{n}_1 - \hat{n}_2)$

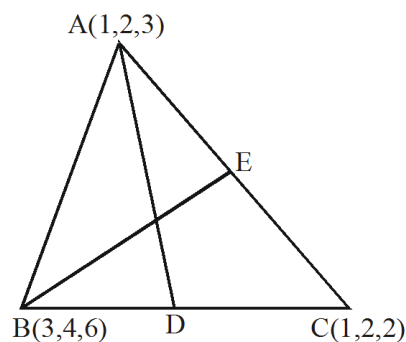
$$\Rightarrow \text{Equation of planes} \Rightarrow \vec{r} \cdot (\hat{n}_1 + \hat{n}_2) = 0$$

$$14x - 5y - 7z = 0 \quad \dots(1)$$

$$\Rightarrow \vec{r} \cdot (\hat{n}_1 - \hat{n}_2) = 0$$

$$2x + 7y - z = 0 \quad \dots(2)$$

6. **Ans (A,C)**



$$D \equiv \left(\frac{\alpha + 1}{2}, 3, \frac{\beta + 6}{2} \right)$$

$$\overrightarrow{AD} = \left(\frac{\alpha - 1}{2} \hat{i} + \hat{j} + \frac{\beta}{2} \hat{k} \right)$$

$$\frac{\alpha - 1}{2} = 1 = \frac{\beta}{2} \Rightarrow (\alpha, \beta) \equiv (3, 2)$$

$$D \equiv (2, 3, 4)$$

median through B :

$$B \equiv (3, 4, 6) \text{ and } E \left(1, 2, \frac{5}{2} \right)$$

$$\frac{x - 3}{4} = \frac{y - 4}{4} = \frac{z - 6}{7}$$

SECTION-I (ii)

7. **Ans (B)**

Let α be the root with $|\alpha| = 1$

$$\text{Clearly } z_1 \alpha^2 + z_2 \alpha + z_3 = 0 \quad \dots(i)$$

$$\text{so } \bar{z}_1 \bar{\alpha}^2 + \bar{z}_2 \bar{\alpha} + \bar{z}_3 = 0$$

$$\Rightarrow \frac{3}{z_1 \alpha^2} + \frac{3}{z_2 \alpha} + \frac{3}{z_3} = 0$$

$$\Rightarrow \frac{1}{z_3} \alpha^2 + \frac{1}{z_2} \alpha + \frac{1}{z_1} = 0 \quad \dots(ii)$$

Eliminating α from (i) and (ii), we get

$$z_2^2 = z_1 z_3$$

8. **Ans (C)**

$$\sum |z_p - z_i|^2 = 3|z_p|^2 + 3|z_1|^2$$

$$= 6|z_1|^2 = 6 \times 3 = 18$$

9. **Ans (A,B,D)**

Point B lies one on plane P_1 . So, point C is the mirror image of point B about the plane P_2 .

$$\text{So, } \frac{\alpha - 2}{3} = \frac{\beta}{-2} = \frac{\gamma + 2}{-2} = K \text{ (say)}$$

$$\Rightarrow C(\alpha, \beta, \gamma) \equiv (3K + 2, -2K, -2K - 2)$$

Now, mid-point of BC will lie on P_2 .

$$\frac{9K + 12}{2} + 2K = 2(-K - 2) + 7 = 0$$

$$\Rightarrow K = -2 \Rightarrow C(-4, 4, 2)$$

Now, D.R. of normal of the plane perpendicular to P_1 & P_2 is given by

$$\begin{aligned} & (2\hat{i} - 3\hat{j} + 6\hat{k}) \times (3\hat{i} - 2\hat{j} - 2\hat{k}) \\ &= (18\hat{i} + 22\hat{j} + 5\hat{k}) \end{aligned}$$

So, required plane is

$$18(x + 4) + 22(y - 4) + 5(z - 2) = 0$$

$$\Rightarrow 18x + 22y + 5z = 26$$

10. **Ans (A,C)**

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SECTION-II

1. Ans (0.66 or 0.67)

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} \hat{a} \cdot \hat{a} & \hat{a} \cdot \hat{b} & \hat{a} \cdot \hat{c} \\ \hat{b} \cdot \hat{a} & \hat{b} \cdot \hat{b} & \hat{b} \cdot \hat{c} \\ \hat{c} \cdot \hat{a} & \hat{c} \cdot \hat{b} & \hat{c} \cdot \hat{c} \end{vmatrix}$$

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{vmatrix}$$

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = (1 - \cos \theta)^2 (1 + 2 \cos \theta) \geq 0$$

$$\cos \theta \geq -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3}$$

2. Ans (31.00)

$$\because BA = AB^2$$

Pre-multiply by B, we get

$$B^2A = (BA)B^2 = (AB^2)B^2 = AB^4$$

Again pre-multiply by B

$$B^3A = (BA)B^4 = (AB^2)B^4$$

$$\Rightarrow B^3A = AB^6 \text{ In general } B^m A = AB^{2m} \dots (1)$$

again $BA = AB^2$ post multiply by A^4

$$\Rightarrow BA^5 = AB^2A^4 \Rightarrow B = AB^2A^4$$

$$\therefore B = A(B^2A)A^3 = A(AB^4)A^3 \text{ (using (1))}$$

$$\therefore B = A^2(B^4A)A^2 = A^2(AB^8)A^2$$

$$\Rightarrow B = A^3(B^8A)A = A^3(AB^{16})A$$

$$\Rightarrow B = A^4(B^{16}A) = A^4 \cdot AB^{32} \Rightarrow B = B^{32}$$

Pre-multiply by B^{-1} , we get $B^{31} = I$

3. Ans (64.00)

$$A = \begin{bmatrix} \sqrt{3} & 1 & 0 \\ 1 & -\sqrt{3} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A^T = \begin{bmatrix} \sqrt{3} & 1 & 0 \\ 1 & -\sqrt{3} & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\Rightarrow AA^T = 4I$$

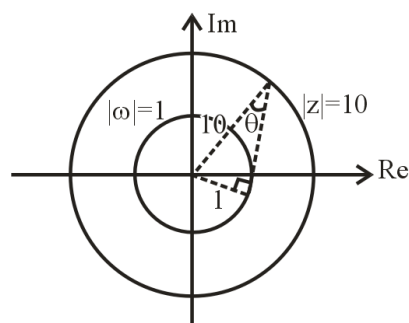
$$\Rightarrow A^T = 4A^{-1} = \frac{4 \cdot \text{adj}A}{|A|} = \frac{4 \cdot \text{adj}A}{-8}$$

$$\Rightarrow 2A^T + \text{adj}A = O$$

$$\therefore d = |2A^T + \text{adj}A + AA^T|$$

$$\Rightarrow d = |4I| = 4^3 |I| = 64$$

4. Ans (99.00)



$$\theta = \arg\left(\frac{w-z}{z}\right)$$

$$\sin \theta = \frac{1}{10}$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{99}}$$

$$\Rightarrow \tan^2 \theta = \frac{1}{99}$$

5. Ans (5.83)

$$P = P(R \ W \ R) + R(W \ R \ W)$$

$$= \frac{2}{5} \times \frac{4}{7} \times \frac{3}{5} + \frac{3}{5} \times \frac{2}{7} \times \frac{1}{5}$$

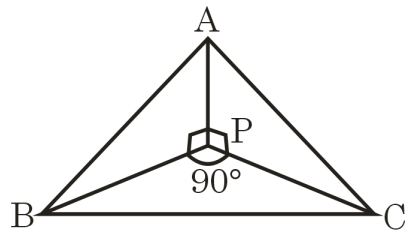
$$= \frac{6}{35}$$

6. **Ans (162.00)**

Let $AP = x$, $PC = y$ & $PB = z$

$$AB = \sqrt{x^2 + z^2}; \quad AC = \sqrt{x^2 + y^2} \quad \&$$

$$BC = \sqrt{y^2 + z^2}$$



$$S = x + y + z + \sqrt{x^2 + y^2} + \sqrt{y^2 + z^2} + \sqrt{z^2 + x^2}$$

Use $AM \geq GM$

$$\Rightarrow x + y + z \geq 3(xyz)^{1/3} \quad \dots(1)$$

$$\sqrt{x^2 + y^2} \geq \sqrt{2xy} \quad \dots(2)$$

$$\sqrt{y^2 + z^2} \geq \sqrt{2yz} \quad \dots(3)$$

$$\sqrt{z^2 + x^2} \geq \sqrt{2zx} \quad \dots(4)$$

$$(1) + (2) + (3) + (4)$$

$$\Rightarrow S \geq 3(xyz)^{1/3} + \sqrt{2} (\sqrt{xy} + \sqrt{yz} + \sqrt{zx})$$

$$\Rightarrow S \geq 3(1 + \sqrt{2})(xyz)^{1/3}$$

$$\Rightarrow V = \frac{1}{6}xyz$$

7. **Ans (3.00)**

$$\vec{OA} = \vec{a}, \vec{OB} = \vec{b} \text{ \& } \vec{OC} = \vec{c}$$

$$\vec{a} + 2\vec{b} + 3\vec{c} = 0$$

$$\vec{a} = -2\vec{b} - 3\vec{c}$$

now proceed

8. **Ans (0.37 or 0.38)**

$$P(T) = \frac{3}{4}$$

$$P(6) = \frac{3}{4} \cdot \frac{1}{6} + \frac{1}{4} \cdot \frac{5}{6} = \frac{1}{3}$$

$$P = \frac{\frac{3}{4} \cdot \frac{1}{6}}{\frac{1}{3}} = \frac{\frac{1}{8}}{\frac{1}{3}} = \frac{3}{8}$$